

Pneumonia Report

491915

Introduction

The aim of this report is to summarise pneumonia patients and their admissions from the MIMIC-III (Medical Information Mart for Intensive Care) database and to present and interpret the seasonal change in admissions as well as the difference in the distribution of the population by age in comparison to the population in the US 2022 census data.

Pneumonia Patients Summary

Characteristics	Patients = 6437 ¹
Age (years)	
0-17	51 (0.8%)
18-59	2,112 (33%)
60+	4,274 (66%)
Diabetes status	
No	4,679 (73%)
Yes	1,758 (27%)
Vaccine status	
0	5,988 (93%)
1+	449 (7.0%)
Gender	
Female	2,854 (44%)
Male	3,583 (56%)
Ethnicity	
Asian	160 (2.5%)
Black/African American	580 (9.0%)
Hispanic/Latino	49 (0.8%)
Other/Mixed	329 (5.1%)
Unknown/Not Answered	655 (10%)
White	4,664 (72%)
¹ n (%)	

A table to show the summary of pneumonia patients identified in the MIMIC database. The patient data included here correspond to each patient's first admission. In some cases further admissions saw a change in the age, diabetes

status or vaccination status fields, but the change was insignificant. 23 patients saw changes in their age category, 74 were diagnosed with diabetes since their last admission and 7 had had a vaccination.

Patient Admissions Summary

Characteristics	Age (years)			Gender	
	0-17 Admissions = 51 ¹	18-59 Admissions = 2408 ¹	60+ Admissions = 4906 ¹	Female Admissions = 3267 ¹	Male Admissions = 4098 ¹
Length of Stay (ICU)	51 (52)	10 (11)	8 (11)	9 (12)	9 (12)
Admissions without ICU stay	0	11	24	16	19
Length of Stay (Hospital)	52 (51)	18 (17)	15 (14)	16 (15)	17 (17)
Death Status					
Alive	50 (98%)	2,078 (86%)	3,776 (77%)	2,640 (81%)	3,264 (80%)
Deceased	1 (2.0%)	330 (14%)	1,130 (23%)	627 (19%)	834 (20%)

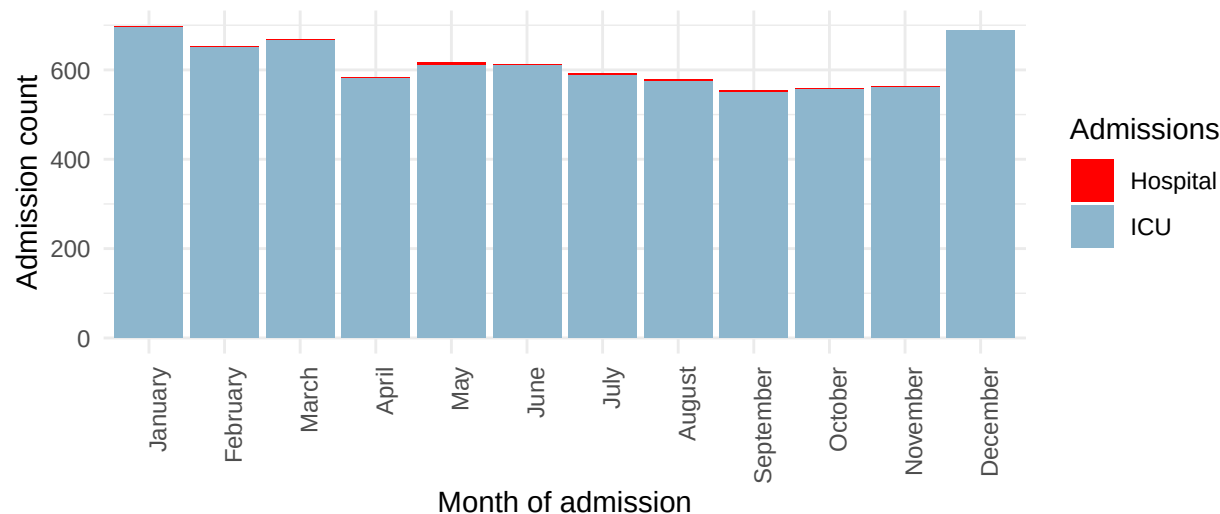
¹Mean (SD); n (%)

Characteristics	Diabetes Status		Vaccination Status	
	No Admissions = 5282 ¹	Yes Admissions = 2083 ¹	0 Admissions = 6834 ¹	1+ Admissions = 531 ¹
Length of Stay (ICU)	9 (13)	9 (10)	9 (12)	8 (12)
Admissions without ICU stay	24	11	28	7
Length of Stay (Hospital)	17 (17)	15 (14)	16 (16)	15 (15)
Death Status				
Alive	4,192 (79%)	1,712 (82%)	5,453 (80%)	451 (85%)
Deceased	1,090 (21%)	371 (18%)	1,381 (20%)	80 (15%)

¹Mean (SD); n (%)

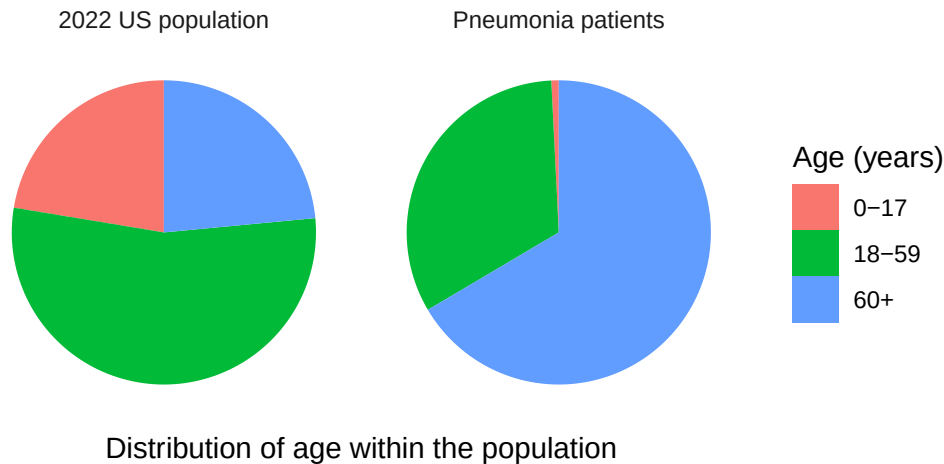
This table displays the summary of admissions (stratified by age, gender, diabetes diagnosis and vaccination status) in terms of the length of stay in the hospital and ICU and also includes a report on the status of death in each stratification. Both length of stay rows display the mean stay in days with the standard deviation in brackets. The amount of patients without an ICU stay is also shown.

Pneumonia Admissions per Month



A bar chart to show the seasonal change in pneumonia hospital admissions and the ICU stays. The bars are layered with the ICU admission count in blue on top of the hospital admission count in red.

2022 US Population Comparison



Pie charts comparing the population distributions, stratified by age, of the pneumonia patients in the MIMIC database and the US 2022 census data.

Interpretation and Discussion

Throughout the report, it appears that people within the 60+ year age category are the most likely group to be hospitalised with pneumonia. This is shown most clearly in the pie charts comparing the pneumonia patients to the US population as it is the only age group that is over-represented. That said, despite having relatively few hospitalisations, the 0-17 year age group experienced some of the longest length of stays on average. So in the interest of reducing the burden on hospitals, vaccination rollouts should target these two age groups first.

Minimal differences were observed in gender and diabetes diagnosis status. The non-vaccinated group had a 5% higher death rate than the vaccinated group, indicating that a higher vaccine uptake could reduce the amount of admissions resulting in death.

Judging by the bar chart the pneumonia admissions peak in the winter months, December through to March, suggesting a link to seasonal flu.